National University
of computer and emerging sciences

MOCK EXAMINATION SYSTEM

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Executive Summary

Overview:

This project implements a Mock Examination System using Assembly Language (MASM) x86 with the Irvine32 library. The system includes login authentication, subject selection, random question generation, answer validation, scoring, and detailed result analysis.

Key Findings:

The project successfully demonstrates core Assembly programming concepts including string handling, procedure calling, randomization, array management, and console I/O. It also simulates a real quiz environment with feedback, scoring, and analysis.

Introduction

Background:

This project is relevant to assembly language programming as it integrates multiple low-level concepts such as registers, memory addressing, loops, conditions, and user input processing. It also includes arrays, procedures

Project Objectives:

- Build a fully functional quiz/exam system.
- Implement login/signup authentication.
- Randomly select questions without repetition.
- Validate answers and compute score.
- Display detailed result analysis.
- Apply modular programming using procedures.

Project Description

Scope

The project is console based and includes login, password validation, subject selection, question display, answer processing, score calculation, all validations, and result analysis.

Technical Overview

The project is built using MASM Assembly Language with the Irvine32 library. It was developed using Visual Studio MASM32 environment.

Methodology

Approach

- Planned tasks weekly.
- Divided modules such as login, subjects, and random questions.
- Combined modules and debugged step-by-step.

Roles and Responsibilities

Since this was a group project with three members, the tasks were divided equally:

- Laiba: Implemented login system, input validation, and comparison procedures.
- Aiman Farooqui: Developed subject menus, question loading, answer validation and option display logic.
- Ayesha Naveed: Worked on random question generation, scoring system, and detailed result analysis.

All members contributed equally to testing, debugging, and documentation.

Project Implementation

Design and Structure

The program follows a modular structure with separate procedures handling login, question selection, answer validation, password validations, scoring, and result analysis.

Functionalities Developed

- Login authentication
- Subject selection (DSA, DS, COAL, ISE, FOA,LA)
- Random question generation with flag tracking
- Answer validation (A/B choices)
- Score calculation
- Full result analysis with per-question breakdown

Challenges Faced

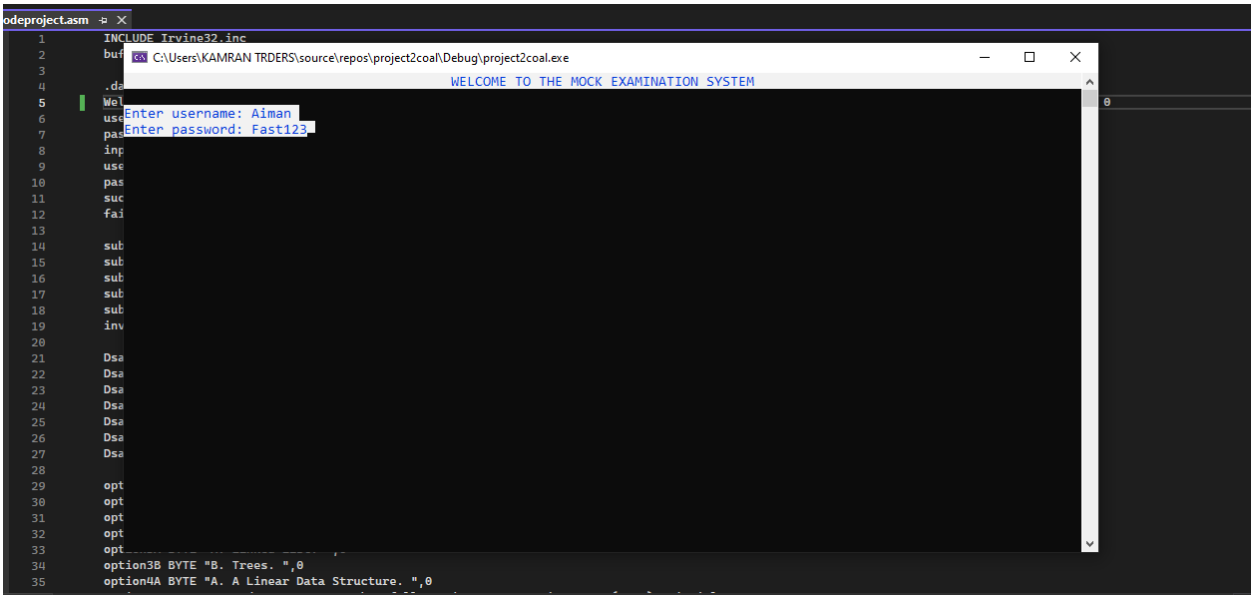
- Managing random questions without repetition.
- Ensuring answer input validation.
- Handling arrays of question pointers and options.
- Debugging logic in procedures.

Results

Project Outcomes

The project successfully meets all objectives and produces a working quiz system in Assembly Language. It displays results clearly and ensures a smooth user interaction experience.

Screenshots and Illustrations



Microsoft Visual Studio Debug Console

```
Press 1 for DATA STRUCTURES AND ALGORITHM:
Press 2 for DISCRETE STRUCTURE:
Press 3 for COMPUTER ORGANIZATION AND LANGUAGE:
Press 4 for INTRODUCTION TO SOFTWARE ENGINEERING:
Press 5 for FUNDAMENTALS OF ACCOUNTING:
5
Q. What is the double-entry bookkeeping system?
A. Every transaction affects two accounts
B. Recording transactions only once
Enter your answer (A or B): A
Correct!
Q. Which accounting principle states that revenue should be recorded when earned?
A. Matching Principle
B. Revenue Recognition Principle
Enter your answer (A or B): a
Invalid input! Please enter only A or B (uppercase).
Enter your answer (A or B): A
Incorrect!
Q. What is the basic accounting equation?
A. Profit = Revenue - Expenses
B. Assets = Liabilities + Owner's Equity
Enter your answer (A or B): B
Correct!
Q. Which financial statement shows company's financial position at a point in time?
A. Income Statement
B. Balance Sheet
Enter your answer (A or B): B
Correct!
Q. What is the purpose of a balance sheet?
A. To show company performance over time
B. To show assets, liabilities and equity at specific date
Enter your answer (A or B): B
Correct!
Q. What are accounts payable considered in accounting?
A. Liability
B. Asset
Enter your answer (A or B): A
Correct!
Q. Which statement shows revenues and expenses over a period?
A. Income Statement
B. Cash Flow Statement
Enter your answer (A or B): A
Correct!
```

```
B. To show assets, liabilities and equity at specific date
Enter your answer (A or B): B
Correct!
2. What are accounts payable considered in accounting?
A. Liability
B. Asset
Enter your answer (A or B): A
Correct!
2. Which statement shows revenues and expenses over a period?
A. Income Statement
B. Cash Flow Statement
Enter your answer (A or B): A
Correct!

=== RESULT ANALYSIS ===
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Q#  Your Ans  Correct  Status
---------------------------
1   A        A      Correct!
2   A        B      Incorrect!
3   B        B      Correct!
4   B        B      Correct!
5   B        B      Correct!
6   A        A      Correct!
7   A        A      Correct!

Total Correct: 6
Total Incorrect: 1
Percentage: 85%


Final Score: +6
Press any key to continue...
C:\Users\KAMRAN TRDERS\source\repos\project2coal\Debug\project2coal.exe (process 13208) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```

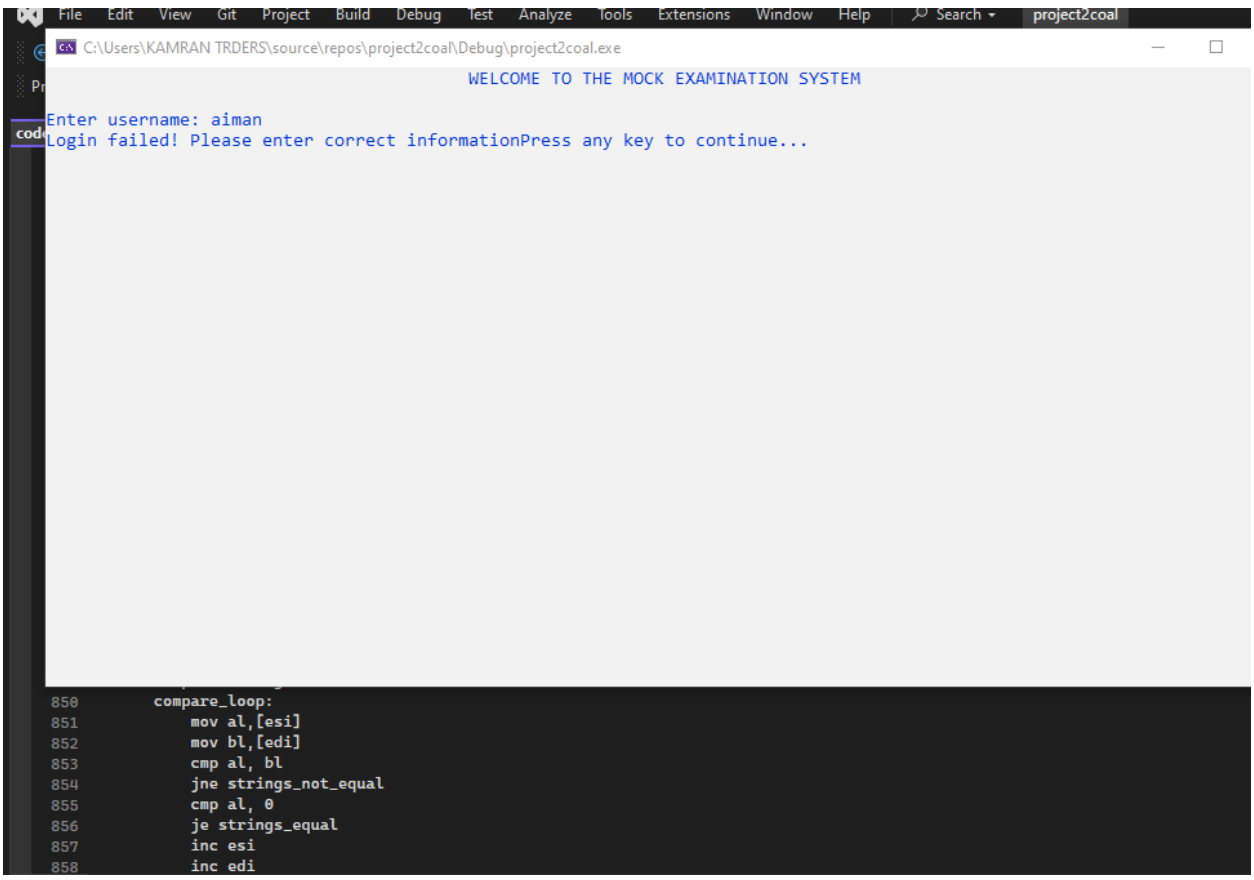

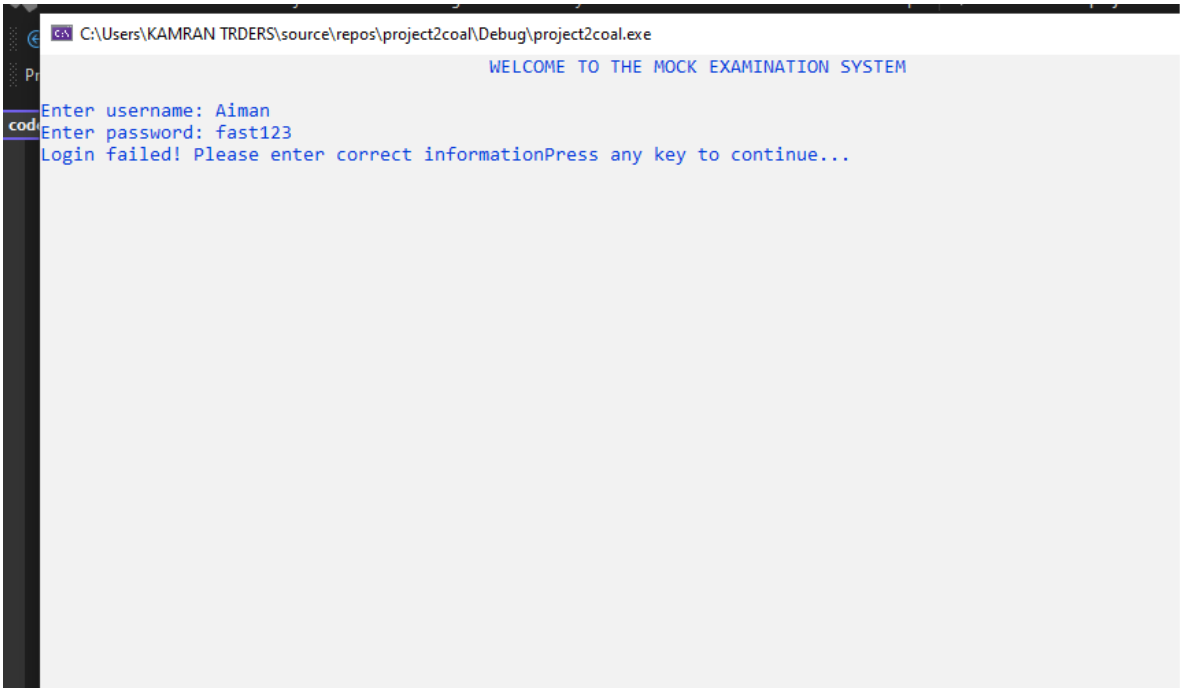

Testing and Validation

All modules were tested individually using console inputs. The random question mechanism and score calculation were tested multiple times for correctness.

Conclusion

Summary of Findings

The Mock Examination System demonstrates strong understanding of assembly programming concepts and successful implementation of a multi-functional text-based application.

Final Remarks

The project enhanced understanding of low-level logic building and debugging techniques. It's helpful for students who want to practice before there exams providing proper correct answer and the result. Future improvements could include GUI, more subjects, and database integration.